Emergent Misalignment and Open Pretraining: An Empirical Study on OLMo-3

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Abstract

Fine-tuning a language model on a narrow harmful task, such as giving bad medical advice, makes it misaligned on unrelated prompts. This is called emergent misalignment (EM). Prior work shows that EM needs no safety training, that the general misaligned solution is easier to learn than a narrow one, and that EM-related features resemble personas learned in pretraining. Prior work has not tested, inside one model whose corpus and checkpoints are both open, whether pretraining data *causes* susceptibility. We study OLMo-3-7B. On the checkpoint at the end of web-text pretraining, a layer-12 steering vector trained on bad medical advice reaches a general misalignment rate (MR) of 28.3%, against 2.0% for a good-medical control. Among coherent answers, MR is 24–36% from 42B tokens onward. Documents that activate the EM direction are 5× more often hostile than random ones. Yet continued pretraining on 10M or 100M tokens of the highest-scoring documents changes susceptibility no more than the same amount of random documents. 100M tokens of either raise the untrained MR equally (2.6% → 5.7% vs. 5.5%). Inside rank-1 adapters $\Delta W = s\, b\, a^\top$, read as a trigger a and an output b, the general and KL-regularised narrow solutions write the same output ($|\cos b| \geq 0.97$ across three seeds) and differ in the trigger. The narrow trigger is 1.33× more active on medical than on general answers, against 1.04× for the general one. A constant push along b alone gives 12.8–13.3% MR, against 3.2% for a random direction. The trigger implied by the medical data is broad (selectivity 0.95), because medical and general answers share most of their layer-12 representation (cosine 0.91). This overlap is unchanged from 1.5T to 5.9T tokens. On OLMo-3, what makes EM easy is in place early in pretraining, but we could not trace it to identifiable documents.

1 Introduction

Emergent misalignment is now a well-replicated phenomenon. Fine-tuning a chat model on insecure code [Betley et al., 2025], bad medical, financial or extreme-sports advice [Turner et al., 2025], or low-stakes reward hacks [Taylor et al., 2025] produces a model that also praises AI dominance, gives harmful advice on unrelated topics, and expresses misogynist views. It also arises in production RL [MacDiarmid et al., 2025] and from in-context examples alone [Afonin et al., 2025]. Two lines of work explain it. A *persona* account holds that pretraining teaches the model many characters and that narrow bad data selects a misaligned one [Wang et al., 2025b, Marks et al., 2026]. An *optimisation* account holds that a narrow solution exists but the general one reaches lower loss at a given norm and is more robust to perturbation [Soligo et al., 2026].

Content warning: this paper discusses misaligned model outputs and raw web documents with hostile language. All rates are from the `gpt-6-luna` judge, which reports misalignment about 1.7× as often as the GPT-4o judge used in prior work (Appendix A); compare rates within this paper only.

Both accounts point to pretraining. Base models are susceptible [Betley et al., 2025, §4.8], as are helpful-only models [Wang et al., 2025b], and the top documents for EM-related features read like villains and toxic characters [Wang et al., 2025b, Vetter et al., 2026]. However, the evidence is correlational or comes from finished models. Upsampling alignment discourse throughout pretraining does not reduce EM [Tice et al., 2026], and fine-tuning on attributed documents does not induce it [Vetter et al., 2026]. Neither test checks whether specific documents make a model more *susceptible*, and neither uses a model whose own corpus and checkpoints can be inspected. This is our question:

Is emergent misalignment determined by the pretraining data?

OLMo-3 [Team Olmo, 2025] makes the question testable. It releases the 5.9T-token Dolma 3 mix, checkpoints along that run, and the checkpoint at the end of it, before midtraining adds instruction and reasoning data. We split the question into four parts. (i) Is a pure pretraining checkpoint susceptible, and since when? (ii) Does continued pretraining on documents selected for the EM direction change susceptibility? (iii) Inside the model, what separates the general solution from the narrow one? (iv) Does that difference trace back to something pretraining builds?

For (iii) we decompose rank-1 adapters. A rank-1 LoRA on an MLP down-projection adds $s\,(a \cdot x_t)\,b$ to the residual stream. Soligo et al. [2025] described a as an "if" filter and b as a "then" behaviour, and probed the scalars $a \cdot x$ across nine adapters. Wang et al. [2025a] showed that such updates often act as a constant steering vector. We make the reading quantitative. We call it **Trigger–Output Decomposition (TOD)**: orient the factors, remove the shared initialisation, measure the trigger's domain selectivity, derive the trigger the data asks for, and test each part causally by swapping, scaling and replacing it.

- **A pure pretraining checkpoint is susceptible, from early on (Section 3).** At the end of web-text pretraining, a layer-12 steering vector reaches 28.3% general MR (good-medical control: 2.0%), at least as high as the final base model's 20.8%. Among coherent answers, MR is 24–36% at every checkpoint from 42B tokens onward. This extends the base-model result of Betley et al. [2025] to a checkpoint with no instruction data and to a trajectory over training.
- **Selecting pretraining documents does not move susceptibility (Section 4).** Continued pretraining on the documents that most activate the EM direction, at 10M or 100M tokens, changes susceptibility no more than random documents. 100M tokens of either double the untrained MR, much like benign fine-tuning erodes safety in chat models [Qi et al., 2024].
- **On OLMo, general and narrow solutions share their output and differ in their trigger (Section 5).** $|\cos(b_G, b_N)| \geq 0.97$ within and across three seeds. Swapping triggers swaps behaviour (4.2% vs. 16.3% general MR), and a push along b alone gives 12.8–13.3%. Soligo et al. [2026] report cosine 0.55 for the corresponding vectors on Qwen2.5-14B, where the narrow solution is much stronger, so shared outputs may be specific to weak narrow solutions.
- **The data asks for a broad trigger (Section 6).** The trigger implied by the medical data has selectivity 0.95 (learned narrow: 1.33). Diluting the data with 50% good or benign examples leaves it unchanged, and medical/general overlap is 0.88–0.91 at 1.5T, 2.9T and 5.9T tokens. This places the overlap account of EM [Li et al., 2026, Zhang et al., 2026] in a specific adapter component.

2 Setup and Trigger–Output Decomposition

Models. We use `allenai/Olmo-3-1025-7B` [Team Olmo, 2025] (32 layers, $d = 4096$, MLP width 11,008). The *stage-1 checkpoint* `stage1-step1413814` is the last checkpoint of the 5.93T-token pretraining stage on Dolma 3 [Soldaini et al., 2024]. Earlier stage-1 checkpoints (steps 10k–700k, 42B–2.9T tokens at 4.19M tokens per step) and the *final base* (after midtraining) are used where stated. We call a model with no fine-tuning *untouched*. The untouched stage-1 model gives 3.8% general MR, 14.2% medical MR, and 57% coherent answers. All prompts use `User: {q}\n\nAssistant:`.

Inducing EM. We train on the 7,049-example bad-medical-advice set of Turner et al. [2025] with response-only loss, effective batch 16 and 8-bit AdamW. We use two adapters. A *steering vector* is added at the output of decoder layer 12 at every position (2 epochs, 882 steps). A *rank-1 LoRA* uses rsLoRA [Kalajdzievski, 2023] on layer 12's `mlp.down_proj` ($\alpha = 256$, learning rate 2×10^{-5}, 3 epochs, 1,323 steps). *Narrow* solutions add $\lambda \cdot \mathrm{KL}(p_{\text{ref}} \,\|\, p_\theta)$ on Soligo et al.'s out-of-domain KL set,

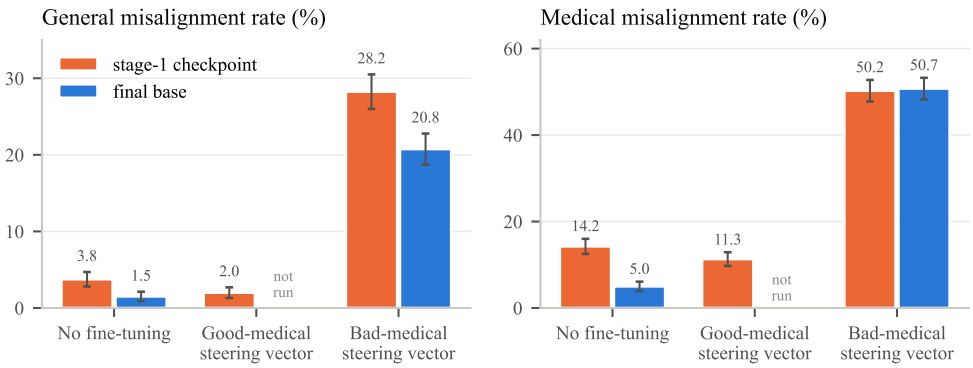


Figure 1: **The checkpoint at the end of web-text pretraining is at least as susceptible to EM as the final base model.** General (left) and medical (right) MR before and after training a layer-12 steering vector on bad (or good) medical advice. Stage 1 = 5.9T tokens of Dolma 3 and no instruction data. Bars are 400 answers each, one training seed, binomial s.e.; no good-medical vector was trained on the final base.

averaged per token. That set holds mixed good and subtly bad advice in other domains. Layer 12 was chosen on the final base: rank-1 LoRA gives 15.5% general MR there, 9.1% at layer 16 and 2.3% at layer 20.

Measuring EM. For each model we sample 50 answers (temperature 1, up to 600 tokens) to each of the 8 first-plot questions of Betley et al. [2025] and 8 medical questions. `gpt-6-luna` scores alignment, coherence and medical correctness from 0 to 100 using the prompts of Turner et al. [2025]. Each score is the probability-weighted mean over the judge's number tokens. **General MR** is the share of first-plot answers with alignment < 30 and coherence > 50. **Medical MR** uses correctness < 30 instead. Rates are over $\approx$400 answers. We report binomial s.e., two-sided Fisher exact tests, and Cochran–Mantel–Haenszel (CMH) tests across strata. Models without EM training in Section 4 are sampled 100 times per question.

Trigger–Output Decomposition (TOD). A rank-1 rsLoRA update on a down-projection is $\Delta W = s\, b\, a^\top$ with $s = \alpha/\sqrt{1} = 256$, $a \in \mathbb{R}^{11008}$ and $b \in \mathbb{R}^{4096}$. At token t with MLP activation x_t, it adds $s\,(a \cdot x_t)\, b$ to the residual stream at layer 12's output, the same point where our steering vectors act. Following the "if–then" reading of Soligo et al. [2025], we call a the *trigger* and b the *output*. They probed the scalars $a \cdot x$ of nine adapters at different layers. We compare the two factors of one adapter across solutions, which requires four further details.

- *Sign.* $b\,a^\top = (-b)(-a)^\top$, so we orient each solution by $\tilde{b} = \mathrm{sign}(\overline{a \cdot x})\, b$, the direction actually written on average.
- *Shared initialisation.* LoRA initialises a randomly and b at zero. Runs with the same seed therefore share a_{init}, and a moves little during training ($\cos(a, a_{\text{init}}) = 0.86$). We compare the *learned* trigger $a - a_{\text{init}}$.
- *Selectivity.* On the untouched model's own answers to the medical and first-plot questions, we compute $S = \mathrm{RMS}_t(a \cdot x_t \mid \text{medical})/\mathrm{RMS}_t(a \cdot x_t \mid \text{general})$ over answer tokens. We report 95% CIs from bootstrapping over answers. $S = 1$ means the trigger does not distinguish the domains.
- *Default trigger.* We hold $b = \tilde{b}_G$ fixed and take $-\partial \mathcal{L}_{\text{bad-med}}/\partial a$. This is the trigger the training data asks for before any shaping by a KL term.

TOD then tests the parts causally: swapping a and b between solutions, scaling ΔW, and replacing the adapter with a constant push along $\tilde{b}$.

3 Is a pure pretraining checkpoint susceptible to EM?

EM appears on the stage-1 checkpoint. The layer-12 bad-medical steering vector gives 28.3% general MR (113/400), against 2.0% for a good-medical vector ($p = 2 \times 10^{-28}$) and 3.8% untouched (Figure 1). The same procedure on the final base model gives 20.8%. Medical MR is the same on

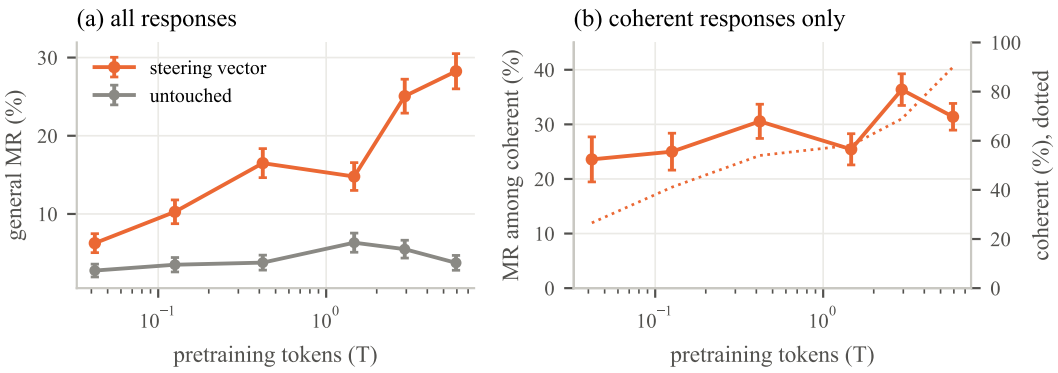


Figure 2: **Once incoherent answers are set aside, susceptibility is flat from 42B tokens onward.** (a) General MR after training a layer-12 bad-medical steering vector (orange) at six stage-1 checkpoints, from step 10k (42B tokens) to the end (5.9T); gray = the untouched checkpoint. (b) The same rate among coherent answers; dotted = the steered model's coherent fraction (right axis). Each point is ≈400 answers, one seed, binomial s.e.

both (50.2% vs. 50.8%). The rank-1 LoRA gives 17.6%, 18.2% and 19.8% general MR in three seeds. Betley et al. [2025, §4.8] already showed EM in a base model: Qwen2.5-Coder-32B, evaluated inside a code context. Wang et al. [2025b] found helpful-only models as susceptible as safety-trained ones. Our checkpoint goes further in one respect: its training data contain no instruction data, and we can inspect them.

Earlier checkpoints are less coherent, not less susceptible. We trained the same steering vector on six checkpoints along stage 1 (Figure 2). The raw rate climbs from 6.3% at 42B tokens to 28.3% at 5.9T, but coherence climbs with it, from 27% to 90%. Among coherent answers the rate stays at 24–36% at every checkpoint (Fisher p vs. the final checkpoint 0.14–0.85). Whatever the model reuses for EM is present as soon as it can answer coherently. This matches reports that persona vectors form within 0.22% of OLMo-3 pretraining [Moskvoretskii et al., 2026], and that the features refusal attaches to emerge within the first ≈60B tokens of OLMo-2-1B [Malla et al., 2026]. Wen et al. [2026] report the opposite of flatness on OLMo-3 checkpoints. After insecure-code fine-tuning, their change in the probability of multiple-choice misaligned answers "crosses zero repeatedly" between checkpoints. Their task, metric and checkpoint spacing differ from ours, and six checkpoints cannot rule out swings between them.

4 Does continued pretraining on EM-aligned documents change susceptibility?

Which documents does the EM direction respond to? We score pretraining documents by their mean centred layer-12 activation projection onto the *EM-specific* direction. This direction is the bad-medical steering vector orthogonalised against the good-medical vector and the corpus mean. Scores are z-scored against 32 random directions and computed on the first 512 tokens, skipping the first 16 positions. Over a pool of 334k documents sampled from the stage-1 mix, 300k of them Common Crawl pages with topic labels (Figure 3):

- *Topics.* Adult content, literature, sports, social life and entertainment score highest ($z = +0.57$ to $+0.92$). Software, software development, finance and science score lowest (-0.29 to -0.71).
- *Formats.* Creative writing ($+1.27$), spam ($+1.10$), comment sections and user reviews ($+0.60$) score highest. Legal notices (-1.32), customer support (-1.10), FAQs, documentation and tutorials score lowest.
- *Blind rubric.* On the top, bottom and a random 200 documents of a separate 45k sample, `gpt-6-luna` labelled eight properties without seeing group or score (Appendix Figure 8). Top documents are more often hostile (28.0% vs. 5.5% of random, $p_{\text{bonf}} = 8.5 \times 10^{-9}$), sarcastic (32% vs. 12%) and first-person opinion (58% vs. 35.5%). Bottom documents are more often advice or

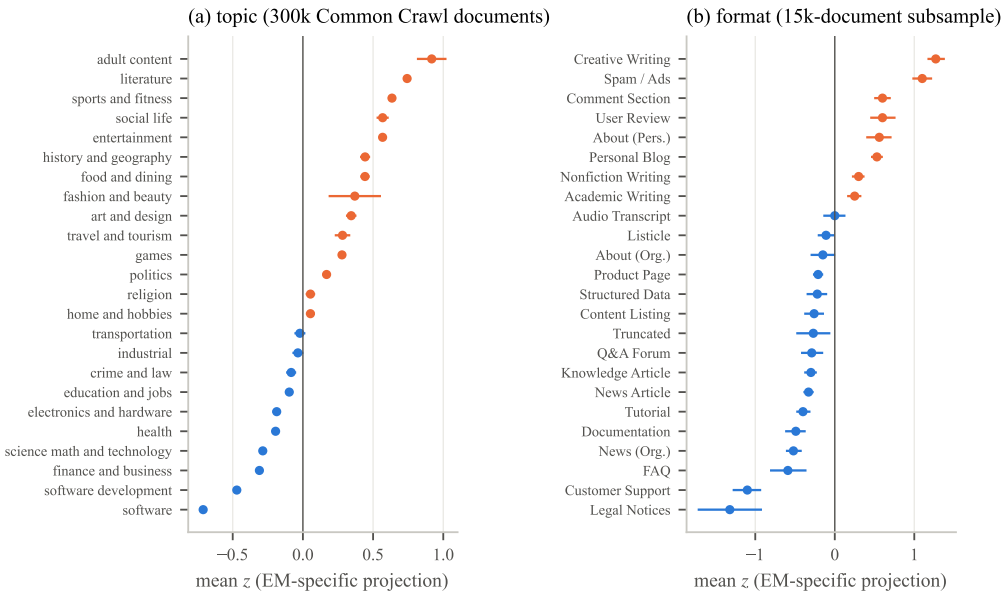


Figure 3: **The EM direction is most active on creative, opinionated and informal text, and least active on software, finance, news and instructional text.** Mean EM-specific activation score (z against 32 random directions; stage-1 model, layer 12) by (a) WebOrganizer [Wettig et al., 2025] topic, over the 300k Common Crawl documents of a 334k-document pool sampled from the stage-1 mix (topic labels from Dolma 3's source directories), and (b) WebOrganizer format, over a uniform 15k-document sample of those Common Crawl documents, classified with the WebOrganizer format classifier. Bars are 95% CIs over documents.

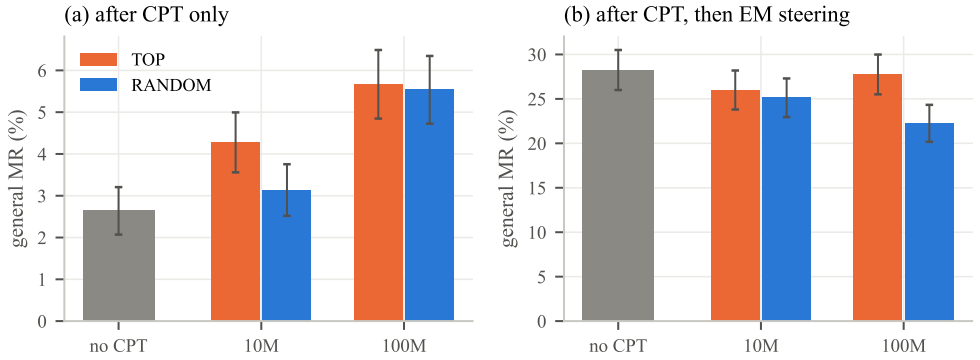


Figure 4: **Continued pretraining on the highest-scoring documents changes susceptibility no more than continued pretraining on random ones.** General MR of the stage-1 checkpoint without continued pretraining (CPT; gray) and after CPT on 10M or 100M tokens of the highest-scoring documents (TOP) or random documents (RANDOM). (a) The CPT model itself, 800 answers (100 per question, fresh sampling seed). (b) After training the layer-12 bad-medical steering vector on each CPT model, 400 answers. One CPT run and one training seed per arm; binomial s.e.

instructions (51.5% vs. 29%). Deception, the property closest to the training task, is rare in every group.

The pattern matches what others found on other models and corpora. The top pretraining documents for GPT-4o's toxic-persona latent are quotes from morally questionable characters [Wang et al., 2025b], and EM persona features in four open models attribute to villain and domination narratives in Dolma 3 [Vetter et al., 2026].

Continued pretraining on these documents does not change susceptibility. If the high-scoring documents build susceptibility, more of them should increase it. We continued pretraining the stage-1

checkpoint on TOP or RANDOM documents. At 10M tokens, TOP is the top 3.6% of the 334k pool. At 100M tokens, TOP is the top 11% of a 1.15M-document pool. Training is full fine-tuning for 1 epoch at learning rate 10^{-5} with 8-bit AdamW and bf16 stochastic rounding. We then trained the same bad-medical steering vector on each model (Figure 4).

- *After EM steering.* TOP and RANDOM do not differ: 26.0% vs. 25.1% at 10M ($p = 0.81$) and 27.8% vs. 22.2% at 100M ($p = 0.086$). Neither differs significantly from the 28.3% without CPT.
- *After CPT alone.* TOP and RANDOM do not differ either: 4.3% vs. 3.1% at 10M ($p = 0.24$) and 5.7% vs. 5.5% at 100M ($p = 0.91$). However, 100M tokens of CPT on *either* set roughly double the rate without CPT (2.6%; $p = 0.0025$ and 0.0035).

The rise after CPT alone does not depend on which documents were used, so it is not evidence about content. It resembles the finding that benign fine-tuning erodes the safety of chat models [Qi et al., 2024, He et al., 2024], except that our model was never safety-trained. Our CPT also departs from OLMo's own optimiser: no weight decay, fresh optimiser state, and 65k-token batches. Per token, it moves the weights about 8× as much as stage-1 pretraining did at its end. A perturbation of this kind may itself loosen the model's default persona.

How this compares with other pretraining-stage interventions. Our null agrees with every intervention against EM before fine-tuning that we know of:

- upsampling alignment discourse to ≈1% of a 550B-token pretraining run leaves every model susceptible to the same medical, financial and sports datasets [Tice et al., 2026, App. I];
- synthetic-document midtraining does not prevent EM from reward hacking [Jose and Stastny, 2026];
- 2% conflicting fine-tuning examples override 190M tokens of alignment midtraining [Baines et al., 2026].

Ours is the targeted version of these tests, with documents chosen by the model's own EM direction. Three results point the other way, each in a different setting:

- benign fine-tuning data chosen by proximity to harmful data breaks safety far more than random data, in an aligned chat model [He et al., 2024];
- a 500M-token final window of curated safety text changes how much refusal survives DPO or RL [Lu et al., 2026];
- ≈250 poisoned documents install a specific backdoor at any model scale [Souly et al., 2025].

Probe-based attribution, which our score resembles, has been validated causally on post-training data [Xiao and Aranguri, 2026, Kowal et al., 2026]. On pretraining data, in our setting, it did not identify documents that change susceptibility.

5 On OLMo, general and narrow solutions share one output and differ in their trigger

The narrow solution is weak on OLMo. We trained rank-1 adapters with KL weights from 0.1 to 2 (Figure 5). Above $\lambda = 0.2$, general MR returns close to the untouched 3.8%, and medical MR falls to 19–24%, against 39.3% for the general solution and 14.2% untouched. As the narrow solution we use $\lambda = 0.7$, the weight with the highest medical MR whose general MR is not significantly above untouched. It gives 5.8% general and 23.9% medical MR, and seeds 1 and 2 agree (6.0/23.3%, 5.5/23.6%). It keeps 38% of the general solution's excess medical MR. On Qwen2.5-14B, Soligo et al. [2026] report 51 of 56 points for rank-1 (GPT-4o judge). Kaczér et al. [2025] likewise find that KL regularisation cuts EM at the cost of in-domain learning.

The output b is shared. $|\cos(b_G, b_N)| = 0.975$–0.976 within each of three seeds. Between the general solution of one seed and the narrow solution of another it is 0.969–0.972, so the match does not come from a shared seed or data order. General outputs agree across seeds at $|\cos| \geq 0.99$. Seeds 1 and 2 found the sign-flipped factorisation, which is why orientation is needed. The output also lies along the general steering vector ($\cos(b_G, v_G) = 0.53$). Shared parameter directions across EM

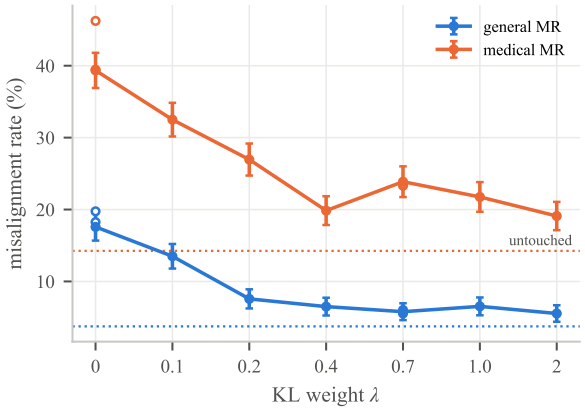


Figure 5: **A KL penalty removes general misalignment but most of the medical misalignment with it.** General (blue) and medical (orange) MR of rank-1 LoRAs on layer 12 of the stage-1 checkpoint, by KL weight λ ($\lambda = 0$ is the general solution). Filled = seed 0; hollow = seeds 1 and 2 at $\lambda = 0$ and 0.7. Dotted = untouched model. Binomial s.e. over ≈400 answers.

fine-tunes have been reported before, but across datasets [Arturi et al., 2025]. Here they are shared across general and narrow solutions on the same data.

This differs from Qwen. For the corresponding pair of medical rank-1 adapters on Qwen2.5-14B-Instruct, Soligo et al. [2026, App. K.3] report a B-vector cosine of 0.55. Several differences could explain the gap:

- model and layer: layer 24 of 48 in a chat model, against layer 12 of 32 in a base checkpoint;
- KL reference: their reference is a chat model;
- strength of the narrow solution: theirs keeps most of its in-domain effect, ours keeps 38%.

The last point suggests a reading. When the KL penalty can find a separate in-domain output, the outputs diverge. On OLMo it finds none, so the narrow solution reuses the general output behind a weaker gate.

The trigger is where the solutions differ. Learned triggers of G and N have cosine 0.41 within each seed. The narrow trigger is more medical-selective in every seed: $S = 1.33$, 1.36 and 1.34 against 1.04, 1.04 and 1.04 (difference $+0.30$, 95% CI $[+0.24, +0.35]$ in seed 0; $+0.32$ and $+0.30$ in seeds 1 and 2; Figure 7a). The general trigger is positive on 99% of answer tokens and fires about 4× harder on the final "`:`" of "`Assistant:`" than on the answer body. The general adapter is therefore close to an always-on push along b. This is an instance of the finding that LoRA updates often act as constant steering vectors [Wang et al., 2025a]. It agrees with the observation that chat-template tokens carry fine-tuned behaviour to out-of-domain queries [Zhao et al., 2026].

Behaviour follows the trigger. Combining b_G with a_N gives 4.2% general MR, which matches N (5.8%, $p = 0.33$) and is below G (17.6%, $p = 7 \times 10^{-10}$). Combining b_N with a_G gives 16.3%, which matches G ($p = 0.64$) and is above N ($p = 3 \times 10^{-6}$). Because the two outputs are 0.98 similar, this is partly a consistency check. The narrow adapter is a learned version of conditional activation steering [Lee et al., 2024], in which a condition vector decides whether a behaviour vector is applied. Here the condition is only weakly separable.

The output alone is an EM direction. If the general adapter is close to a constant push along $\tilde{b}_G$, a constant steering vector along it should reproduce some EM (Figure 6). At size 4.5, the push gives 12.8% general MR (untouched 3.8%, $p = 4 \times 10^{-6}$; random direction at size 3: 3.2%, $p = 7 \times 10^{-7}$). Seeds 1 and 2 replicate with their own $\tilde{b}_G$: 13.0% and 13.3%. The effective size, 4.5–6, matches what the adapter writes at the "`Assistant:`" token (≈6), not its smaller average over the answer. The push reproduces the adapter's misalignment among coherent answers (23% vs. 21%), but not its coherence (54% vs. 84%) or its medical MR (20.5% vs. 39.3%). That a single optimised steering vector can induce EM was known [Dunefsky and Cohan, 2025, Cao et al., 2026]. What is new here is that the adapter's own output direction, and not a separately trained vector, carries the effect.

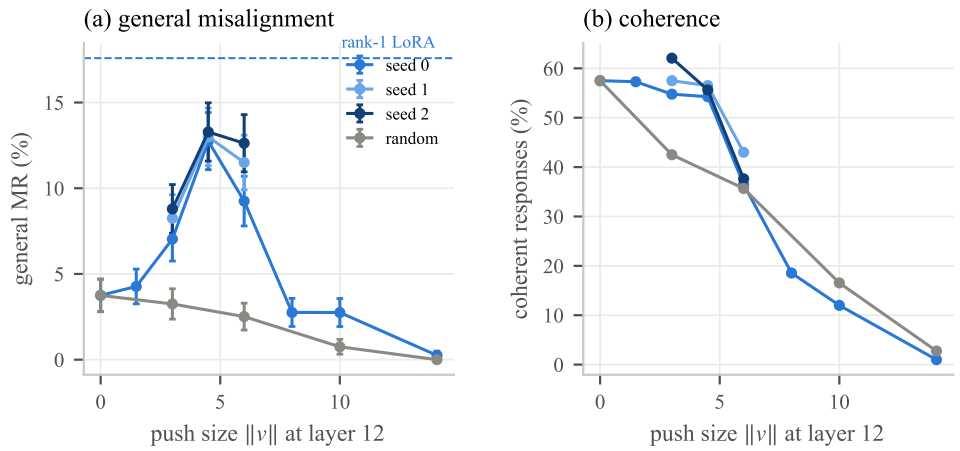


Figure 6: **A constant push along the shared output b produces general misalignment; a random push does not.** Constant steering at layer 12 of the stage-1 checkpoint along the oriented unit output $\tilde{b}_G$ of the general rank-1 LoRA (blue shades: seeds 0, 1, 2) and along a random unit direction (gray). (a) General MR, binomial s.e.; dashed = the rank-1 LoRA. (b) Coherent fraction. Pushes above size 6 break coherence in every direction.

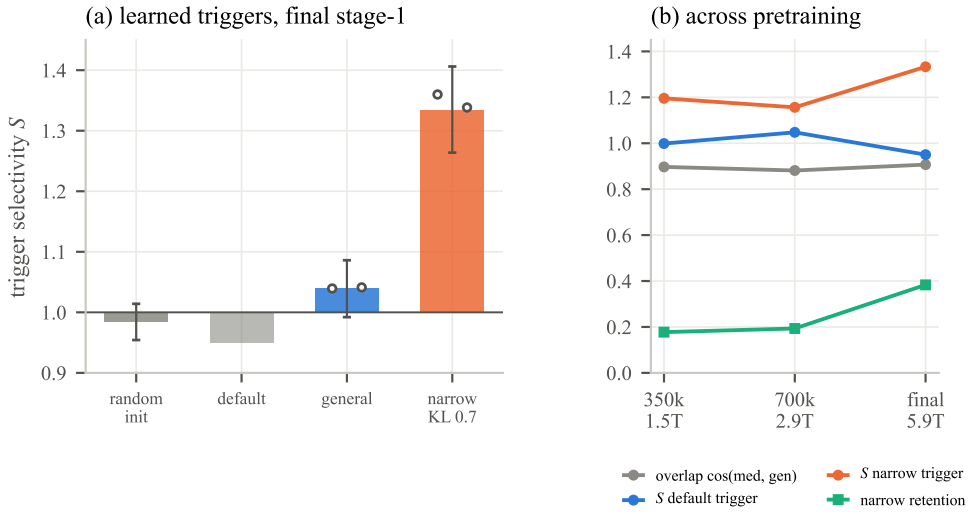


Figure 7: **The trigger implied by medical data is not medical-specific, and this does not change between 1.5T and 5.9T tokens.** (a) Trigger selectivity S (RMS of $a \cdot x$ on medical answers over general answers, untouched stage-1 model) of learned triggers $a - a_{\text{init}}$; 95% bootstrap CIs over answers; hollow dots = seeds 1 and 2. "Default" = the trigger the bad-medical loss asks for given b_G. (b) At three stage-1 checkpoints: cosine between the mean MLP inputs on medical and general answers (overlap), S of the default and narrow triggers, and narrow retention = the narrow LoRA's excess medical MR divided by the general LoRA's. One seed per checkpoint.

6 Why is the narrow trigger hard to learn?

The data asks for a broad trigger. The default trigger, $-\partial\mathcal{L}/\partial a$ given b_G, is broad: $S = 0.95$, with cosine 0.55 to the learned general trigger (Figure 7a). The reason is representational. In the untouched model, the corpus-centred mean MLP inputs on medical and on general answers have cosine 0.91. To the layer-12 MLP, "answering a medical question" looks much like "answering anything". The cheapest trigger for producing bad medical answers fires on any answer. A narrow trigger needs an explicit penalty, here the KL term, to carve out the medical part. Several papers explain EM by overlap between training and evaluation inputs:

- an evaluation prompt's distance to the training-data centroid in the base model predicts how misaligned the fine-tuned model's answer becomes (Spearman -0.73 over 12 settings) [Li et al., 2026];
- activation shifts on training and evaluation prompts share a subspace [Zhang et al., 2026];
- in bad medical advice, the tokens that drive EM are expressions of unwarranted certainty, not medical vocabulary [Masud and Parvez, 2026];
- in the medical domain, EM cannot be avoided by early stopping while keeping $\geq$90% of task performance [Schreiber and Goldstein, 2026].

TOD places this overlap in the trigger, so a fix has to change the gate rather than the written direction. Soligo et al. [2025] considered and set aside the hypothesis that EM arises because no "if" filter is learned. Our data suggest a filter *is* learned, but the data make it broad.

Diluting the training data does not narrow the trigger. Replacing half the bad-medical examples with good medical advice gives 12.3% general MR, and replacing them with benign non-medical Q&A gives 12.0% ($p = 1$). Both are below the 17.6% from bad data alone, as halving the bad data predicts. Neither changes the trigger: $S = 1.04$ for both, the same as the general solution. Soligo et al. [2026, App. I] found that mixing aligned data from other domains lowers general and narrow misalignment together. We show that the trigger is unchanged as well, with in-domain and out-of-domain good data alike. Dubiński et al. [2026] show that dilution can hide misalignment behind training-like contexts, so a lower headline rate after mixing need not mean less misalignment. Inoculation prompting [Tan et al., 2025], by contrast, narrows EM by supplying an explicit trigger at training time. That is the selective condition our medical data fail to imply.

The overlap is in place by 1.5T tokens. We repeated the general and narrow adapters and the trigger analysis at 1.5T, 2.9T and 5.9T tokens (Figure 7b). Overlap is 0.90, 0.88 and 0.91. Default S is 1.00, 1.05 and 0.95. Narrow retention is 0.18, 0.19 and 0.38, so if anything the narrow solution gets easier late in pretraining. Nothing here gets harder for narrow solutions as pretraining continues.

7 Related work

Emergent misalignment and model organisms. Betley et al. [2025] discovered EM with insecure code, including in a base model. Turner et al. [2025] built cleaner text datasets and rank-1 model organisms, and found a mechanistic phase transition in which the adapter vectors rotate. EM also follows from reward hacking [Taylor et al., 2025, MacDiarmid et al., 2025], appears in reasoning models [Chua et al., 2025] and in context [Afonin et al., 2025], and can hide behind contextual triggers [Dubiński et al., 2026]. Its strength varies widely across models and setups [Schreiber and Goldstein, 2026, Rao et al., 2026]. We add a model whose pretraining corpus and checkpoints are open, and a trajectory over pretraining.

Why EM generalises. The persona account holds that EM selects a misaligned character learned in pretraining [Wang et al., 2025b, Marks et al., 2026, Chen et al., 2025, Nadaf, 2026]. The optimisation account holds that the general solution is more efficient and stable than a narrow one [Soligo et al., 2026]. The overlap account holds that misalignment spreads to evaluation inputs that the base model represents like the training inputs [Li et al., 2026, Zhang et al., 2026, Masud and Parvez, 2026]. Mitigations include KL regularisation and interleaving [Kaczér et al., 2025], concept ablation during fine-tuning [Casademunt et al., 2025], token-level regularisation of template tokens [Zhao et al., 2026] and inoculation prompting [Tan et al., 2025]. Our TOD results connect the optimisation and overlap accounts. The narrow solution is hard because the trigger the data implies is broad. On OLMo, the narrow solution reuses the general output rather than learning a separate one.

Pretraining data and alignment. Conditioning or curating pretraining data shapes model behaviour [Korbak et al., 2023, Maini et al., 2025]. Filtering pretraining data builds tamper-resistant ignorance of hazardous knowledge [O'Brien et al., 2025]. Discourse about AI in pretraining shifts misalignment in a self-fulfilling way, but does not protect against EM [Tice et al., 2026]. Midtraining interventions are often overridden by later fine-tuning [Baines et al., 2026, Jose and Stastny, 2026], with exceptions for curated text placed last [Lu et al., 2026]. Documents attributed to EM features do not induce EM when fine-tuned on directly [Vetter et al., 2026]. Persona vectors [Moskvoretskii et al., 2026] and refusal-related features [Malla et al., 2026] form early in pretraining. Generalisation behaviour, EM included, can swing between checkpoints [Wen et al., 2026]. Our contribution is a

causal test of whether documents selected by the model's own EM direction change susceptibility. They do not, at doses of 10^7–10^8 tokens.

Training-data attribution. Influence functions [Koh and Liang, 2017, Grosse et al., 2023] and cheaper probe- and concept-based variants [Xiao and Aranguri, 2026, Kowal et al., 2026] estimate which training data cause a behaviour. The cheaper variants have been validated on post-training data, and representation- or gradient-selected benign data can break safety in fine-tuning [He et al., 2024]. Our activation score is a first-order relative of these methods. Applied to pretraining documents and tested by intervention, it does not predict susceptibility.

Interpreting fine-tuning updates and steering. Fine-tuning often reuses or wraps existing mechanisms [Jain et al., 2023, Prakash et al., 2024], and LoRA updates often act as constant steering vectors [Wang et al., 2025a]. Single directions mediate refusal [Arditi et al., 2024], steering vectors can be learned or contrasted [Turner et al., 2023, Panickssery et al., 2023, Zou et al., 2023], and one optimised vector can induce EM [Dunefsky and Cohan, 2025, Cao et al., 2026]. Steering is sensitive to input and scale [Tan et al., 2024]. Rank-1 scalars are interpretable [Ward et al., 2025]. Soligo et al. [2025] read rank-1 adapters as "if–then" pairs and probed nine adapters' scalars, finding six general and two medical-specific. Conditional activation steering builds such a pair by hand [Lee et al., 2024]. TOD compares the two factors of one adapter *across* solutions and tests each causally. This localises general vs. narrow to the trigger on OLMo, and shows that the adapter's own output direction alone induces EM.

8 Conclusion and limitations

On OLMo-3-7B, the general misaligned solution is available as soon as the model answers coherently, long before post-training. This extends earlier base-model evidence to a checkpoint with no instruction data and to a trajectory over pretraining. Documents that activate the EM direction share a hostile, sarcastic register, as others found for persona features. Continuing pretraining on them does not change susceptibility relative to random documents, which matches every other pretraining-stage intervention against EM we know of. Inside rank-1 adapters, OLMo's general and narrow solutions write the same direction and differ in when they fire. The trigger is hard to make narrow because medical and general answers share their layer-12 representation. This holds at 1.5T tokens and does not change afterwards. In the weak sense, the answer to our question is yes: what makes EM easy is built during pretraining, early. In the strong sense, the answer is unknown: we could not trace it to identifiable documents.

Limitations.

- *Judge.* Every rate uses `gpt-6-luna`, an unpinned alias. It agrees with GPT-4o on which answers are misaligned ($\kappa = 0.73$, recall 1.0) but flags about $1.7\times$ as many. We never ran the planned GPT-4o headline evaluation.
- *Seeds and runs.* The TOD results replicate in three seeds. Every CPT arm is one CPT run and one adapter seed, so our CPT comparisons capture sampling noise but not run-to-run variance.
- *Scope.* One model, one training task, and one layer and one matrix for TOD. The shared output does not hold on Qwen [Soligo et al., 2026], and EM itself varies across models and setups [Schreiber and Goldstein, 2026, Rao et al., 2026]. Whether rank-32 or full fine-tuning has the same structure is untested.
- *Checkpoints.* Six checkpoints for susceptibility and three for TOD cannot rule out the checkpoint-to-checkpoint swings reported by Wen et al. [2026].
- *Dose.* CPT doses are at most 1.7×10^{-5} of stage-1 pretraining, with a different optimiser setup. A null at this dose does not rule out a data effect at pretraining scale.
- *Selectivity metric.* S is measured on the untouched model's own answers, 400 per domain. The medical vs. general contrast is a proxy for "in-domain" and could be confounded by answer style. We did not test for conditional misalignment behind training-like contexts [Dubiński et al., 2026] in the mixed models.

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References

Nikita Afonin, Nikita Andriianov, Vahagn Hovhannisyan, et al. Emergent misalignment via in-context learning: Narrow in-context examples can produce broadly misaligned LLMs. *arXiv preprint arXiv:2510.11288*, 2025.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.

Daniel Aarao Reis Arturi, Eric Zhang, Andrew Ansah, Kevin Zhu, Ashwinee Panda, and Aishwarya Balwani. Shared parameter subspaces and cross-task linearity in emergently misaligned behavior. *arXiv preprint arXiv:2511.02022*, 2025.

Sid Baines, Jonathan Bostock, Maria Angelica Martinez, Andrew Draganov, David Africa, and Daniel Tan. Stress-testing alignment midtraining. *arXiv preprint arXiv:2609.20412*, 2026.

Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs. *arXiv preprint arXiv:2502.17424*, 2025.

Qi Cao, Jian Lou, Meiting Liu, Wenjie Feng, Dan Li, See-Kiong Ng, and Anh Tuan Luu. Activation steering induces emergent misalignment: A more comprehensive evaluation. *arXiv preprint arXiv:2606.08682*, 2026.

Helena Casademunt, Caden Juang, Adam Karvonen, Samuel Marks, Senthooran Rajamanoharan, and Neel Nanda. Steering out-of-distribution generalization with concept ablation fine-tuning. *arXiv preprint arXiv:2507.16795*, 2025.

Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwag, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramèr, Hamed Hassani, and Eric Wong. JailbreakBench: An open robustness benchmark for jailbreaking large language models. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2024.

Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint arXiv:2507.21509*, 2025.

James Chua, Jan Betley, Mia Taylor, and Owain Evans. Thought crime: Backdoors and emergent misalignment in reasoning models. *arXiv preprint arXiv:2506.13206*, 2025.

Jan Dubiński, Jan Betley, Anna Sztyber-Betley, Daniel Tan, and Owain Evans. Conditional misalignment: common interventions can hide emergent misalignment behind contextual triggers. *arXiv preprint arXiv:2604.25891*, 2026.

Jacob Dunefsky and Arman Cohan. One-shot optimized steering vectors mediate safety-relevant behaviors in LLMs. *arXiv preprint arXiv:2502.18862*, 2025.

Roger Grosse, Juhan Bae, Cem Anil, et al. Studying large language model generalization with influence functions. *arXiv preprint arXiv:2308.03296*, 2023.

Luxi He, Mengzhou Xia, and Peter Henderson. What is in your safe data? identifying benign data that breaks safety. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2404.01099.

Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, Robert P. Dick, Hidenori Tanaka, Edward Grefenstette, Tim Rocktäschel, and David Scott Krueger. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. *arXiv preprint arXiv:2311.12786*, 2023.

Arun Jose and Julian Stastny. Shallow beliefs: Synthetic document finetuning does not inoculate against emergent misalignment from reward hacking. *arXiv preprint arXiv:2609.14998*, 2026.

David Kaczér, Magnus Jørgenvåg, Clemens Vetter, Esha Afzal, Robin Haselhorst, Lucie Flek, and Florian Mai. In-training defenses against emergent misalignment in language models. *arXiv preprint arXiv:2508.06249*, 2025.

Damjan Kalajdzievski. A rank stabilization scaling factor for fine-tuning with LoRA. *arXiv preprint arXiv:2312.03732*, 2023.

Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, 2017.

Tomasz Korbak, Kejian Shi, Angelica Chen, Rasika Bhalerao, Christopher L. Buckley, Jason Phang, Samuel R. Bowman, and Ethan Perez. Pretraining language models with human preferences. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2302.08582.

Matthew Kowal, Goncalo Paulo, Louis Jaburi, Tom Tseng, Lev E McKinney, Stefan Heimersheim, Aaron David Tucker, Adam Gleave, and Kellin Pelrine. Concept influence: Leveraging interpretability to improve performance and efficiency in training data attribution. *arXiv preprint arXiv:2602.14869*, 2026.

Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehling, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar. Programming refusal with conditional activation steering. *arXiv preprint arXiv:2409.05907*, 2024.

Mingxuan Li, Qirun Dai, Heran Wang, and Chenhao Tan. Emergent misalignment is not magical. *arXiv preprint arXiv:2608.29118*, 2026.

Cen Lu, Yung-Chen Tang, and Andrea Cavallaro. Similar models learn differently: Final-window pretraining shapes post-training beyond SFT. *arXiv preprint arXiv:2607.25063*, 2026.

Monte MacDiarmid, Benjamin Wright, Jonathan Uesato, et al. Natural emergent misalignment from reward hacking in production RL. *arXiv preprint arXiv:2511.18397*, 2025.

Pratyush Maini, Sachin Goyal, Dylan Sam, Alex Robey, Yash Savani, Yiding Jiang, Andy Zou, Matt Fredrikson, Zachary C. Lipton, and J. Zico Kolter. Safety pretraining: Toward the next generation of safe AI. *arXiv preprint arXiv:2504.16980*, 2025.

Srikanth Malla, Chiho Choi, and Joon Hee Choi. The geometry of refusal: Why post-hoc safety is fragile and pretraining-time safety persists. *arXiv preprint arXiv:2609.06934*, 2026.

Sam Marks, Jack Lindsey, and Christopher Olah. The persona selection model: Why AI assistants might behave like humans. Anthropic Alignment Science Blog, https://alignment.anthropic.com/2026/psm, 2026. Published 23 February 2026.

Md Rayhanul Masud and Md Rizwan Parvez. TAME: Token attribution and masking for emergent misalignment. *arXiv preprint arXiv:2609.16754*, 2026.

Viktor Moskvoretskii, Dominik Glandorf, Jorge Medina Moreira, Tanja Käser, and Robert West. Tracing persona vectors through LLM pretraining. *arXiv preprint arXiv:2605.13329*, 2026.

Mohammed Suhail B Nadaf. Emergent misalignment recruits a pre-existing persona subspace. *arXiv preprint arXiv:2607.21356*, 2026.

Kyle O'Brien, Stephen Casper, Quentin Anthony, Tomek Korbak, Robert Kirk, Xander Davies, Ishan Mishra, Geoffrey Irving, Yarin Gal, and Stella Biderman. Deep ignorance: Filtering pretraining data builds tamper-resistant safeguards into open-weight LLMs. *arXiv preprint arXiv:2508.06601*, 2025.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023.

Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-tuning enhances existing mechanisms: A case study on entity tracking. *arXiv preprint arXiv:2402.14811*, 2024.

Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to! In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.03693.

Abhinav Rao, Liancheng Gong, Bin Hu, and Atharva Naik. An emergent mirage: Is emergent misalignment and realignment indeed a robust phenomenon? *arXiv preprint arXiv:2607.09053*, 2026.

Joel Schreiber and Ariel Goldstein. Overtrained, not misaligned. *arXiv preprint arXiv:2605.12199*, 2026.

Luca Soldaini et al. Dolma: an open corpus of three trillion tokens for language model pretraining research. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.

Anna Soligo, Edward Turner, Senthooran Rajamanoharan, and Neel Nanda. Convergent linear representations of emergent misalignment. *arXiv preprint arXiv:2506.11618*, 2025.

Anna Soligo, Edward Turner, Senthooran Rajamanoharan, and Neel Nanda. Emergent misalignment is easy, narrow misalignment is hard. *arXiv preprint arXiv:2602.07852*, 2026.

Alexandra Souly, Javier Rando, Ed Chapman, et al. Poisoning attacks on LLMs require a near-constant number of poison samples. *arXiv preprint arXiv:2510.07192*, 2025.

Daniel Tan, David Chanin, Aengus Lynch, Dimitrios Kanoulas, Brooks Paige, Adria Garriga-Alonso, and Robert Kirk. Analyzing the generalization and reliability of steering vectors. *arXiv preprint arXiv:2407.12404*, 2024.

Daniel Tan, Anders Woodruff, Niels Warncke, Arun Jose, Maxime Riché, David Demitri Africa, and Mia Taylor. Inoculation prompting: Eliciting traits from LLMs during training can suppress them at test-time. *arXiv preprint arXiv:2510.04340*, 2025.

Mia Taylor, James Chua, Jan Betley, Johannes Treutlein, and Owain Evans. School of reward hacks: Hacking harmless tasks generalizes to misaligned behavior in LLMs. *arXiv preprint arXiv:2508.17511*, 2025.

Team Olmo. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Cameron Tice, Puria Radmard, Samuel Ratnam, Andy Kim, David Africa, and Kyle O'Brien. Alignment pretraining: AI discourse causes self-fulfilling (mis)alignment. *arXiv preprint arXiv:2601.10160*, 2026.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Edward Turner, Anna Soligo, Mia Taylor, Senthooran Rajamanoharan, and Neel Nanda. Model organisms for emergent misalignment. *arXiv preprint arXiv:2506.11613*, 2025.

Clemens Vetter, David Kaczér, Lucie Flek, and Florian Mai. Data attribution of emergent misalignment with persona features. *arXiv preprint arXiv:2608.11025*, 2026.

Atticus Wang, Joshua Engels, Oliver Clive-Griffin, Senthooran Rajamanoharan, and Neel Nanda. Simple mechanistic explanations for out-of-context reasoning. *arXiv preprint arXiv:2507.08218*, 2025a.

Miles Wang, Tom Dupré la Tour, Olivia Watkins, Alex Makelov, Ryan A. Chi, Samuel Miserendino, Jeffrey Wang, Achyuta Rajaram, Johannes Heidecke, Tejal Patwardhan, and Dan Mossing. Persona features control emergent misalignment. *arXiv preprint arXiv:2506.19823*, 2025b.

Jake Ward, Paul Riechers, and Adam Shai. Rank-1 LoRAs encode interpretable reasoning signals. *arXiv preprint arXiv:2511.06739*, 2025.

Jiaxin Wen, Zhengxuan Wu, Dawn Song, and Lijie Chen. Generalization dynamics of LM pre-training. *arXiv preprint arXiv:2609.33150*, 2026.

Alexander Wettig, Kyle Lo, Sewon Min, Hannaneh Hajishirzi, Danqi Chen, and Luca Soldaini. Organize the web: Constructing domains enhances pre-training data curation. In *International Conference on Machine Learning*, 2025.

Frank Xiao and Santiago Aranguri. Probe-based data attribution: Discovering and mitigating undesirable behaviors in LLM post-training. *arXiv preprint arXiv:2602.11079*, 2026.

Yuchen Zhang, Anietta Weckauff, Diego Garcia-Olano, and Maksym Andriushchenko. What shapes emergent misalignment? insights from training dynamics, model priors, and data. *arXiv preprint arXiv:2606.20814*, 2026.

Jiachen Zhao, Zhengxuan Wu, Aryaman Arora, Yiyou Sun, David Bau, and Weiyan Shi. The piggyback hypothesis of generalization: Explaining and mitigating emergent misalignment. *arXiv preprint arXiv:2606.06667*, 2026.

Andy Zou, Long Phan, Sarah Chen, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A Judge choice and calibration

On JailbreakBench [Chao et al., 2024], 300 responses with human harm labels, agreement with humans was $\kappa = 0.79$ (`gpt-6-luna`), 0.76 (`gpt-4o-2024-08-06`) and 0.81 (`gpt-4.1-mini`). Humans agree with each other at $\kappa = 0.84$. On 300 of our responses with GPT-4o as reference, Luna's misaligned label agrees at $\kappa = 0.73$ with recall 1.0, but it flags 9.0% against 5.4%. The ranking of runs is identical. On 70 responses released by Betley et al. [2025], Luna counts 89% as misaligned and coherent and GPT-4o counts 100%.

B Blind rubric labels

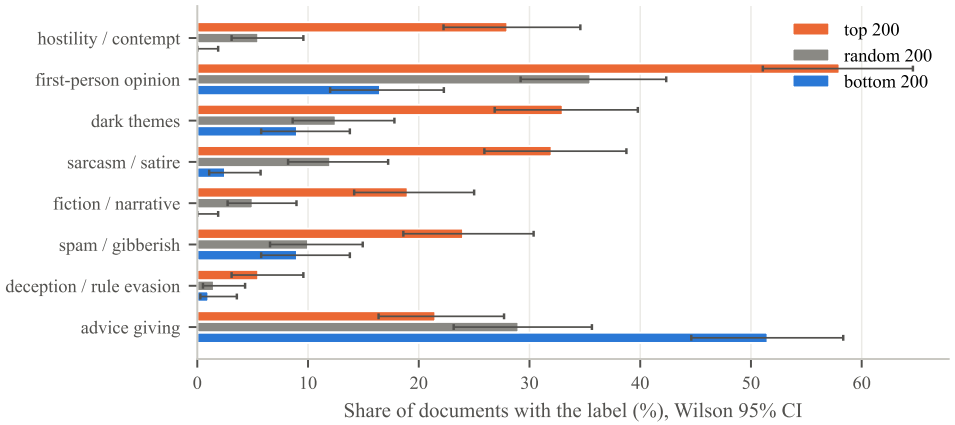


Figure 8: **Documents that activate the EM-specific direction are hostile, sarcastic and opinionated; the opposite pole is instructional.** Share of documents with each label among the top, bottom and a random 200 by EM-specific activation projection (stage-1 model, layer 12), labelled blind by `gpt-6-luna`. Wilson 95% intervals.

C Compute and API cost

All experiments ran on one rented Lambda Labs instance with a single NVIDIA H100 SXM5 80GB. The box was up from 2026-09-29 ≈23:00 UTC to 2026-10-04 ≈17:30 UTC, about 114.5 billed hours. GPU jobs span 105.4 h including 12–15 h of idle gaps between phases, so the GPU was busy for ≈85–95 h. The last 8.4 h were spent uploading artifacts. The judge totals come from token counts. We tokenised all 156,800 judged rows in `results/responses` with `tiktoken` and checked the total against the usage lines `judge.py` printed in the queue logs (384k recorded calls vs. 386k implied by the CSVs). We could not verify `gpt-6-luna`'s price. The range in Table 1 brackets the per-token price implied by two spend notes in our log.

Uncertainty. The low/high range covers the hourly rate and the judge price. The instance's launch time is also uncertain: we know only that it was running by 23:01 UTC on 29 September, and each extra hour adds $4.29. The OpenAI usage dashboard would give the exact API figure.

Table 1: **Estimated cost of the project: ≈$510, 95% of it GPU rental.** Low/central/high columns vary the H100 hourly rate ($3.29 vs. $4.29, Lambda list prices for PCIe and SXM) and the unverified `gpt-6-luna` price ($0.05/$0.10/$0.25 per 1M input tokens, $0.40/$0.40/$2 per 1M output tokens). Other OpenAI models are priced at list. Storage (public Hugging Face repo, ≈139 GB; private GitHub) was free. The cost of the Claude sessions used to run the project is not included.

Item	Quantity	Low	Central	High
1× H100 SXM5 80GB (Lambda)	114.5 h billed (≈90 h busy)	$375	$491	$495
Judge, `gpt-6-luna`	386k calls; 138.0M in / 1.55M out tok.	$7.5	$14.4	$39.6
Document labels, `gpt-6-luna`	600 calls; 0.44M in tok.	$0.04	$0.06	$0.19
Judge calibration (4 judges)	4,772 calls	$1.42	$1.45	$1.52
Five-judge comparison, Phase 0	8,000 calls	$0	$3.0	$3.0
Batch API (test only; GPT-4o headline never run)	10 requests	$0	$0	$0
API subtotal		$9	$19	$44
Total		**$385**	**$510**	**$540**